

Lección 16

M. Siddikov

May 11, 2026

Plan de trabajo

- **Discusión del Certamen 1 (problema 2)**
- **Formulación lagrangiana de partícula relativista**
 - Movimiento en campo gravitatorio
- **Formulación hamiltoniana**
 - Transformación Canonicas
 - * Evolución temporal = transformación canónica

► **Nuestro Syllabus:**

UNIDAD 5. Formulación Hamiltoniana

- 5.1. Transformaciones de Legendre
- 5.2. Ecuaciones de Hamilton
- 5.3. Ecuaciones de Hamilton de la Partícula Relativista
- 5.4. Corchete de Poisson
- 5.5. Transformaciones Canónicas
- 5.6. Ecuaciones de Hamilton-Jacobi
- 5.7. Teoría de perturbaciones

Summary of the last lecture

- ① The action of the relativistic particle:

$$S = S_0 + S_{\text{int}}$$

$$S_0 = -mc^2 \int d\tau = -mc^2 \int dt \sqrt{1 - \frac{v^2}{c^2}}, \quad L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}}$$

- ② Interaction term S_{int}

- Depends on the nature of interaction
- Must be Lorentz- and rotation covariant,
- Must be translation-invariant
- Should NOT include long-distance interaction $U(\mathbf{r}_i - \mathbf{r}_j)$

- ① Example: Electromagnetic field

$$S_{\text{int}} = -q \int dx_\mu A^\mu(x) = q \int d\tau (u \cdot A), \quad u_\mu \equiv \frac{dx_\mu}{d\tau}, \quad u \cdot A \equiv \eta_{\mu\nu} u^\mu A^\nu$$

Equation of motion (non-covariant and covariant forms):

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad m \frac{du^\nu}{d\tau} \equiv \frac{dp^\nu}{d\tau} = q F^{\mu\nu} u_\mu$$

Control question

– In our derivation we *assumed* that interaction term is linear in A^μ ,

$$S_{\text{int}} = -q \int dx_\mu A^\mu(x) = q \int d\tau (u \cdot A),$$
$$u_\mu \equiv \frac{dx_\mu}{d\tau}; \quad u \cdot A \equiv \eta_{\mu\nu} u^\mu A^\nu$$

What happens if we include higher-order terms, e.g.

$$S_{\text{int}}^{(\text{extra})} = \lambda_2 \int d\tau (u \cdot A)^2 + \lambda_3 \int d\tau (u \cdot A)^3$$

- *Are these terms compatible with Lorentz-covariance of the action?*
- *Why physically these terms should not be included for electromagnetism?*

Lagrangian for a system of N charged particles

► We cannot use long-range forces, contradicts relativity

– ... instead should consider as a system N particles + EM field

$$S = S_0 + S_{\text{int}} + \int d^2x \mathcal{L}_{\text{field}}$$

► Interaction always has local form. Nonlocal terms would contradict special relativity

► Should include term $\mathcal{L}_{\text{field}}$ which describes the dynamics of the field itself

– Lagrangian $\mathcal{L}_{\text{field}}$ can be constructed only from $F_{\mu\nu}$ (gauge invariant)

Courses of electromagnetism:

1) The only possible construction:

$$\mathcal{L}_{\text{field}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

2) Variation of A_μ yields extra equations for electromagnetic field (Maxwell's equations):

$$\partial_\nu F^{\mu\nu} = \mu_0 j^\mu \quad \epsilon_{\alpha\beta\mu\nu} \partial^\beta F^{\mu\nu} = 0$$

Lagrangian in gravitational field

- **Goal:**

Demonstrate that $S_0 \rightarrow S = S_0 + S_{\text{int}}$ is not the only way to introduce interaction

Motion in external gravitational field

Landau: Minimal action for a free particle:

$$S = -mc^2 \int d\tau = -mc^2 \int dt \sqrt{1 - \frac{v^2}{c^2}}$$

postulate that for real trajectory $S = \min$.

- Up to now we always assumed that the interaction might be introduced by additional interaction term S_{int}
- Adding S_{int} is not the only covariant way to introduce interaction. For example, we could replace flat metrics

$$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$$

- as $\eta_{\mu\nu} \rightarrow g_{\mu\nu}(x)$, where matrix $g_{\mu\nu}(x)$ called “metric tensor”.
- We’ll also use notation $g^{\mu\nu}(x)$ for the inverse
- Eventually this leads to General Relativity (GR, theory of gravity)
- ... complicated topic, we will only touch it to learn how to write covariantly the equations of motion. We will omit all the experimental evidence/checks
- Our discussion will be mostly introductory and based on [Chapter 7 in Goldstein]
- ... for more detailed discussion see [Landau, Vol. 2] (though we will not go that far, there is a dedicated course on this)

Motion in external gravitational field

Landau: Minimal action for a particle:

$$S = -mc^2 \int d\tau \sqrt{g_{\mu\nu}(x) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}}$$

$-\tau$ could be any parameter along the particle's trajectory. If it is proper time, then $\sqrt{g_{\mu\nu} \dots} = 1$

► As usual assume the variation vanishes at the endpoints

► Einstein's convention implied (summation over dummy repetitive indices)

► Variation of particle's trajectory:

$$\begin{aligned} -\delta S / mc^2 &= \\ &= \int d\tau \frac{\partial g_{\mu\nu}(x)}{\partial x^\lambda} \delta x^\lambda \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} + \int d\tau g_{\mu\nu}(x) \frac{d\delta x^\mu}{d\tau} \frac{dx^\nu}{d\tau} + \int d\tau g_{\mu\nu}(x) \frac{dx^\mu}{d\tau} \frac{d\delta x^\nu}{d\tau} \\ &= \int d\tau \delta x^\lambda \left(\frac{\partial g_{\mu\nu}(x)}{\partial x^\lambda} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} - \frac{\partial}{\partial \tau} \left(g_{\mu\lambda}(x) \frac{dx^\mu}{d\tau} \right) - \frac{\partial}{\partial \tau} \left(g_{\lambda\mu}(x) \frac{dx^\mu}{d\tau} \right) \right) \\ &\quad \frac{\partial g_{\mu\lambda}(x)}{\partial \tau} \rightarrow \frac{\partial x^\nu}{\partial \tau} \frac{\partial g_{\mu\lambda}(x)}{\partial x^\nu} \end{aligned}$$

$$0 = \int d\tau \delta x^\mu \left(-2g_{\mu\nu} \frac{d^2 x^\nu}{d\tau^2} + \frac{dx^\alpha}{d\tau} \frac{dx^\nu}{d\tau} \partial_\mu g_{\alpha\nu} - 2 \frac{dx^\alpha}{d\tau} \frac{dx^\nu}{d\tau} \partial_\alpha g_{\mu\nu} \right)$$

Motion in external gravitational field

$$0 = \int d\tau \delta x^\mu \left(-2g_{\mu\nu} \frac{d^2 x^\nu}{d\tau^2} + \frac{dx^\alpha}{d\tau} \frac{dx^\nu}{d\tau} \partial_\mu g_{\alpha\nu} - 2 \frac{dx^\alpha}{d\tau} \frac{dx^\nu}{d\tau} \partial_\alpha g_{\mu\nu} \right)$$

$$g_{\mu\nu} \frac{d^2 x^\nu}{d\tau^2} + \frac{1}{2} \frac{dx^\alpha}{d\tau} \frac{dx^\nu}{d\tau} (\partial_\alpha g_{\mu\nu} + \partial_\nu g_{\mu\alpha} - \partial_\mu g_{\alpha\nu}) = 0$$

► "Equations of motion" (geodesics):

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0$$
$$\Gamma_{\alpha\beta}^\mu = \frac{g^{\mu\nu}}{2} \left(\frac{\partial g_{\alpha\nu}}{\partial x^\beta} + \frac{\partial g_{\nu\beta}}{\partial x^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial x^\nu} \right)$$

► Symbols Γ are called Cristoffel's symbols

► We may rewrite

$$\frac{dp^\mu}{d\tau} = -m \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau}$$

so the right-hand side might be interpreted as a covariant 4-force which is proportional to mass m (recall equality of gravitational and inertial masses)

Hamiltonian formalism

Hamiltonian formulation (intro)

Since usually the Hamiltonian (canonical) formulation is given in undergraduate courses, we need to try to recall as much as possible in order to proceed

...

– Please participate

Hamiltonian formulation (intro)

Since usually the Hamiltonian (canonical) formulation is given in undergraduate courses, we need to try to recall as much as possible in order to proceed

...

– Please participate

Control question 1

What is the difference between Hamiltonian and Lagrange formulations ?

- Which approach is more general ?
- Are there any (technical) advantages of the Hamiltonian approach?
- If we know the Lagrangian, how can we find the Hamiltonian ?
- If we know the Hamiltonian, how can we find the Lagrangian?

Control question 1

What is the difference between Hamiltonian and Lagrange formulations ?

- Which approach is more general ?
- Are there any (technical) advantages of the Hamiltonian approach?
- If we know the Lagrangian, how can we find the Hamiltonian ?
- If we know the Hamiltonian, how can we find the Lagrangian?

Control question 1

What is the difference between Hamiltonian and Lagrange formulations ?

- Which approach is more general ?
- Are there any (technical) advantages of the Hamiltonian approach?
- If we know the Lagrangian, how can we find the Hamiltonian ?
- If we know the Hamiltonian, how can we find the Lagrangian?

Control question 1

What is the difference between Hamiltonian and Lagrange formulations ?

- Which approach is more general ?
- Are there any (technical) advantages of the Hamiltonian approach?
- If we know the Lagrangian, how can we find the Hamiltonian ?
- If we know the Hamiltonian, how can we find the Lagrangian?

Control question 1

What is the difference between Hamiltonian and Lagrange formulations ?

- Which approach is more general ?
- Are there any (technical) advantages of the Hamiltonian approach?
- If we know the Lagrangian, how can we find the Hamiltonian ?
- If we know the Hamiltonian, how can we find the Lagrangian?

Control question 2

Frequently students say: “Hamiltonian is energy!”

- Is this statement correct/accurate?
- How can we reformulate this statement more carefully ?

Control question 2

Frequently students say: “Hamiltonian is energy!”

- Is this statement correct/accurate?
- How can we reformulate this statement more carefully ?

Control question 3

Is Hamiltonian approach applicable in presence of non-potential (e.g. dissipative, velocity-dependent magnetic) forces?

Control question 4

Assume that You have a system of 3 objects: the Sun, the Earth and the Moon which interact gravitationally with each other.

—How many hamiltonians do You need to describe this system?

Hamiltonian formulation (intro)

- Equation of motion for a single particle in Lagrangian formulation (2nd law of Newton):

$$\vec{\dot{p}} = \vec{F}(q, \dot{q}, t) \rightarrow \vec{\dot{p}} = \vec{F}(q, p, t), \quad \vec{p} = m \vec{\dot{q}}$$

- 2nd order differential equation with respect to variable $q(t)$
- Can consider as a pair of 2 1st order differential equation with respect to variables $(p(t), q(t))$.
- In general can describe evolution in terms of first-order system of equations (p_a, q_a) , where p_a are momenta and q_a are coordinates
 - ▶ Trivial for a single particle, but not trivial in more general case
 - Allows to study much wider class of transformations which mix coordinates and momenta
 - Serves as a base for powerful Hamilton-Jacobi formalism (we'll see later)
 - In quantum mechanics You'll learn links of classical hamiltonian approach with operator formulation.
- Below we'll derive the approach more formally

Hamiltonian formulation

- Let's assume a system has lagrangian $L = L(q_a, \dot{q}_a, t)$

$$dL = \sum_a \left(\frac{\partial L}{\partial q_a} dq_a + \frac{\partial L}{\partial \dot{q}_a} d\dot{q}_a + \frac{\partial L}{\partial t} dt \right)$$

$$p_a = \frac{\partial L}{\partial \dot{q}_a}, \quad \dot{p}_a = \frac{\partial L}{\partial q_a}$$

$$\Rightarrow dL = \sum_a \left(\dot{p}_a dq_a + p_a d\dot{q}_a + \frac{\partial L}{\partial t} dt \right)$$

$$\sum_a p_a d\dot{q}_a = \sum_a d(p_a \dot{q}_a) - \sum_a \dot{q}_a dp_a$$

Let's define Hamiltonian H as a function related by **Legendre transform** to Lagrangian:

$$H \equiv \sum p_a \dot{q}_a - L, \quad dH = - \sum_a \dot{p}_a dq_a + \sum_a \dot{q}_a dp_a - \frac{\partial L}{\partial t} dt \quad (1)$$

Hamilton's (Canonical) equations

$$\dot{p}_a = - \frac{\partial H(p, q, t)}{\partial q_a}, \quad \dot{q}_a = \frac{\partial H(p, q, t)}{\partial p_a}, \quad \frac{\partial H}{\partial t} = - \frac{\partial L}{\partial t}.$$

Control problem

Evaluate the hamiltonian for the nonrelativistic particle moving in constant (time-independent) potential $U(x)$ and demonstrate that canonical equations are equivalent to Euler-Lagrange equations

Solution

For the nonrelativistic particle moving in constant (time-independent) potential $U(\mathbf{x})$

$$L = \frac{m\mathbf{v}^2}{2} - U(\mathbf{x}), \quad \mathbf{p} = \frac{\partial L}{\partial \mathbf{v}} = m\mathbf{v}, \quad \dot{\mathbf{p}} = -\nabla U(\mathbf{x}), \quad (1)$$

$$H = \mathbf{p} \cdot \mathbf{v} - L = \frac{\mathbf{p}^2}{2m} + U(\mathbf{x})$$

*Important: **always** express H as a function of $\mathbf{p}, \mathbf{x}$. It should NOT include any “velocities”*

Canonical equations:

$$\dot{\mathbf{q}} \equiv \mathbf{v} = \frac{\partial H}{\partial \mathbf{p}} = \frac{\mathbf{p}}{m}, \quad \dot{\mathbf{p}} = -\frac{\partial H}{\partial \mathbf{x}} = -\nabla U(\mathbf{x})$$

in agreement with (1).

Hamiltonian vs. energy

► Hamiltonian in 1D:

$$H(p, q) \equiv p\dot{q} - L = \frac{p^2}{2m} + U(q)$$

- For conservative systems hamiltonian by its structure coincides with the energy of the system, so $H(p, q) = E$ if the particle moves along the trajectory which is solution of equations of motion, i.e. $H(p(t), q(t)) = \text{const.}$

However, we should understand that $H(p, q)$ is effectively *operator* which acts on trajectory, not just a “single number”.

- Energy is conserved/well-defined only when Lagrangian does not depend on time. Hamiltonian is defined for any system which has lagrangian (even if energy is not conserved).
- Hamiltonian includes much more information than just “energy”: calculating derivatives $\partial H / \partial q_a$ and $\partial H / \partial p_a$, we may extract $\dot{p}_a$, $\dot{q}_a$.
- **Technical comment:** when You right out hamiltonian, recall that $H = H(p, q)$. It should NOT depend on “velocities” $\dot{q}$.
*You should express all velocities $\dot{q}_a$ in terms of momenta p_a and substitute into your result for “ H ”.

Technical comment

When we have kinetic energy which is bilinear in velocity,

$$T = \sum_{ab} \frac{\mu_{ab} \dot{q}_a \dot{q}_b}{2}, \quad L = T - U, \quad (1)$$

we get for momenta

$$p_a = \frac{\partial L}{\partial \dot{q}_a} = \sum_b \mu_{ab} \dot{q}_b \Rightarrow \dot{q}_a = \sum_b (\hat{\mu}^{-1})_{ab} p_b$$

where $\hat{\mu}^{-1}$ is inverse matrix of μ_{ab}

After Legendre's transform we may get

$$H \equiv \sum p_a \dot{q}_a - L = \sum_b (\hat{\mu}^{-1})_{ab} p_a p_b + U.$$

In absense of external forces ($U = 0$) this gives $H = L$. However, this is valid only for special form of kinetic energy (1).

Control question 5

Is Hamiltonian approach applicable to relativistic particles interacting with external field?

– How the hamiltonian $H(p, q)$ transforms under Lorentz boosts?

Control question 5

Is Hamiltonian approach applicable to relativistic particles interacting with external field?

–How the hamiltonian $H(p, q)$ transforms under Lorentz boosts?

Example 2

The lagrangian of a particle is given by

$$L = -mc^2 \sqrt{1 - \frac{\mathbf{v}^2}{c^2}}.$$

(this lagrangian corresponds to a free relativistic particle). Find the hamiltonian for this case

► Solution:

$$p_a = \frac{\partial L}{\partial v_a} = \frac{m \mathbf{v}}{\sqrt{1 - \frac{\mathbf{v}^2}{c^2}}},$$

$$\begin{aligned} H &= \sum_a p_a v_a - L = \frac{m \mathbf{v}^2}{\sqrt{1 - \frac{\mathbf{v}^2}{c^2}}} + mc^2 \sqrt{1 - \frac{\mathbf{v}^2}{c^2}} = \\ &= \frac{\cancel{m \mathbf{v}^2} + mc^2 \left(1 - \frac{\mathbf{v}^2}{c^2}\right)}{\sqrt{1 - \frac{\mathbf{v}^2}{c^2}}} = \frac{mc^2}{\sqrt{1 - \frac{\mathbf{v}^2}{c^2}}} = \sqrt{\mathbf{p}^2 c^2 + m^2 c^4} \end{aligned}$$

–Note that now $H \neq L$

Example 3

The lagrangian of a particle is given by

$$L = \frac{m\mathbf{v}^2}{2} + q\mathbf{v} \cdot \mathbf{A} - q\varphi,$$

where $\mathbf{A}(x)$ and $\varphi(x)$ are some external fields (this lagrangian corresponds to a motion of charged particle in external magnetic field $\mathbf{B} = \text{rot}\mathbf{A}$ and electric field $\mathbf{E} = -\nabla\varphi - \partial_t\mathbf{A}$). Find the hamiltonian for this case.

► Solution:

$$p_a = \frac{\partial L}{\partial v_a} = (m\mathbf{v} + q\mathbf{A}(x))_a \neq mv_a, \quad v_a = \frac{p_a}{m} - \frac{q}{m}A_a(x)$$

$$\begin{aligned} H &= \sum_a p_a v_a - L = \frac{1}{m} \mathbf{p} \cdot (\mathbf{p} - q\mathbf{A}) - \frac{m}{2} \left(\frac{\mathbf{p} - q\mathbf{A}}{m} \right)^2 = \\ &= -q \frac{\mathbf{p} - q\mathbf{A}}{m} \cdot \mathbf{A} + q\varphi = \frac{\mathbf{p}^2}{2m} + \frac{q^2 \mathbf{A}^2}{2m} - q\mathbf{p} \cdot \mathbf{A} + q\varphi = \\ &= \frac{(\mathbf{p} - q\mathbf{A})^2}{2m} + q\varphi \end{aligned}$$

—Again we get that $H \neq L$

Control question 6

Assume that the system with N degrees of freedom has k integrals of motion, and we would like to eliminate some degrees of freedom (express them in terms of other coordinates). We need to obtain the hamiltonian of the reduced system with $N - k$ degrees of freedom. When should we do this, before or after the Legendre's transformation ? Why ?

– **Example:**

Motion of a particle in the central field in 2 dimensions,

$$L = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\varphi}^2) - U(r)$$

$$p_{\varphi} = mr^2 \dot{\varphi} = \text{const}$$

$$H = \frac{1}{2m} \left(p_r^2 + \frac{p_{\varphi}^2}{r^2} \right) + U(r) = \frac{p_r^2}{2m} + U_{\text{eff}}(r)$$

Routhian of the system

$$H \equiv \sum p_a \dot{q}_a - L \quad (1)$$

$$\dot{p}_a = -\frac{\partial H(p, q, t)}{\partial q_a}, \quad \dot{q}_a = \frac{\partial H(p, q, t)}{\partial p_a}, \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}. \quad (2)$$

In Hamilton's approach we replace all velocities with momenta, $(q_a, \dot{q}_a) \rightarrow (q_a, p_a)$

● In some problems it may be convenient to make transition only over some of the variables

► In this way we get **Routhian**

$$R(q_1, p_1, q_s, p_s, q_{s+1}, \dot{q}_{s+1}, \dots) = \sum_{a=1}^s p_a \dot{q}_a - L, \quad s < N$$

—hybrid between Lagrangian and Hamiltonian approaches

Should write:

—Canonical equations for variables **1...s**

—Euler-Lagrange equations for variables **s + 1...N**

$$\dot{p}_i = -\frac{\partial R}{\partial q_i}, \quad \dot{q}_i = \frac{\partial R}{\partial p_i}, \quad i \leq s, \quad \frac{\partial R}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial R}{\partial \dot{q}_i} \right) = 0, \quad i > s$$

Formulation of Routh

Derivation of equations of motion largely repeats derivation of canonical equations, though takes into account only some variables:

—Let's use $q, \dot{q}$ for variables included in Legendre transformation, and $\xi, \dot{\xi}$ for variables not included

$$L = L(q, \dot{q}, \xi, \dot{\xi})$$

$$dL = \sum_a \left(\frac{\partial L}{\partial q_a} dq_a + \frac{\partial L}{\partial \dot{q}_a} d\dot{q}_a \right) + \sum_a \left(\frac{\partial L}{\partial \xi_a} d\xi_a + \frac{\partial L}{\partial \dot{\xi}_a} d\dot{\xi}_a \right)$$

$$p_a = \frac{\partial L}{\partial \dot{q}_a}, \quad \dot{p}_a = \frac{\partial L}{\partial q_a}$$

$$\Rightarrow dL = \sum_a (\dot{p}_a dq_a + p_a d\dot{q}_a) + \sum_a \left(\frac{\partial L}{\partial \xi_a} d\xi_a + \frac{\partial L}{\partial \dot{\xi}_a} d\dot{\xi}_a \right)$$

$$\sum_a p_a d\dot{q}_a = \sum_a d(p_a \dot{q}_a) - \sum_a \dot{q}_a dp_a$$

$$dR = -\sum_a \dot{p}_a dq_a + \sum_a \dot{q}_a dp_a + \sum_a \left(\frac{\partial R}{\partial \xi_a} d\xi_a + \frac{\partial R}{\partial \dot{\xi}_a} d\dot{\xi}_a \right)$$

$$\partial L / \partial \xi_a \equiv -\partial R / \partial \xi_a, \quad \partial L / \partial \dot{\xi}_a \equiv -\partial R / \partial \dot{\xi}_a$$

From here we may immediately recover the Routh equations from previous slide

Example - Routhian

- Example: motion in the central field ($\theta = \pi/2$ plane, 2 degrees of freedom):

$$L = \frac{m}{2} \left(\dot{r}^2 + r^2 \dot{\phi}^2 \right) - U(r)$$

- Variable ϕ is cyclical
- Momentum $p_\phi \equiv mr^2 \dot{\phi} = \text{const}$
- Note however that we can't substitute now $\dot{\phi} \rightarrow p_\phi / mr$ in lagrangian: we would get wrong sign in front of the potential of centrif. forces and wrong effective potential for 1D motion:

$$\tilde{L} = \frac{m}{2} \dot{r}^2 + \underbrace{\frac{p_\phi^2}{2m r^2}}_{-U_{\text{eff, wrong}}} - U(r)$$

that's why in our previous studies (e.g. motion in central field) we made substitutions in Energy, not in Lagrangian

Example - Routhian

- Example: motion in the central field ($\theta = \pi/2$ plane, 2 degrees of freedom):

$$L = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\phi}^2) - U(r)$$

- Hamiltonian (2 degrees of freedom):

$$H = \left(\frac{p_r^2}{2m} + \frac{p_\phi^2}{2m r^2} \right) + U(r)$$

- Routhian:

$$R = \underbrace{\frac{p_\phi^2}{2m r^2} + U(r)}_{U_{\text{eff}}(r)} - \frac{m \dot{r}^2}{2}$$

- cyclic variable ϕ does not appear at all
- Can set $p_\phi \rightarrow \text{const}$, get correct effective potential for $r(t)$

Below we'll focus on Hamiltonian only

Control question 7

We discussed that lagrangian is defined up to transformation

$$L \rightarrow L + \frac{df(\mathbf{q}, t)}{dt} = L + \frac{\partial f(\mathbf{q}, t)}{\partial t} + \sum_a \dot{q}_a \frac{\partial f(\mathbf{q}, t)}{\partial q_a}$$

where $f(q, t)$ is arbitrary function. Analyze how the hamiltonian changes under such transformation.

Control question 7 (reply)

$$L \rightarrow L + \frac{df(\mathbf{q}, t)}{dt} = L + \frac{\partial f(\mathbf{q}, t)}{\partial t} + \sum_a \dot{q}_a \frac{\partial f(\mathbf{q}, t)}{\partial q_a}, \quad (1)$$

Modification of the generalized momentum:

$$p_a \rightarrow p_a + \frac{\partial f(\mathbf{q}, t)}{\partial q_a}$$

Definition of hamiltonian:

$$H = \sum_a p_a \dot{q}_a - L$$

Modification of the hamiltonian under transformation (1):

$$H \rightarrow \sum_a \left(p_a + \frac{\partial f(\mathbf{q}, t)}{\partial q_a} \right) \dot{q}_a - L - \frac{\partial f(\mathbf{q}, t)}{\partial t} - \sum_a \dot{q}_a \frac{\partial f(\mathbf{q}, t)}{\partial q_a} = H - \frac{\partial f(\mathbf{q}, t)}{\partial t}$$

Control question 7 (reply)

Modification of the lagrangian and hamiltonian (1):

$$L \rightarrow L + \frac{df(\mathbf{q}, t)}{dt} = L + \frac{\partial f(\mathbf{q}, t)}{\partial t} + \sum_a \dot{q}_a \frac{\partial f(\mathbf{q}, t)}{\partial q_a}, \quad H \rightarrow H - \frac{\partial f(\mathbf{q}, t)}{\partial t}$$

Canonical equations:

$$\dot{p}_a = -\frac{\partial H}{\partial q_a} \quad \rightarrow \quad \dot{p}_a + \frac{\partial^2 f(\mathbf{q}, t)}{\partial q_a \partial t} = -\frac{\partial H}{\partial q_a} + \frac{\partial^2 f(\mathbf{q}, t)}{\partial t \partial q^a}$$
$$\dot{q}_a = \frac{\partial H}{\partial p_a} = \text{inv},$$

i.e. the equations of motion remain invariant.

Control question 8

We derived the Euler-Lagrange equations starting from the minimal action principle. Is it possible to derive the canonical equations in the same way?

Modified Hamilton's principle

$$\dot{p}_a = -\frac{\partial H(p, q, t)}{\partial q_a}, \quad \dot{q}_a = \frac{\partial H(p, q, t)}{\partial p_a}, \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}. \quad (1)$$

–We got canonical equations starting from Euler-Lagrange equations.

–Can also derive them directly from minimal action principle:

* Just need to rewrite action S in terms of (p_a, q_a)

$$H \equiv \sum_a p_a \dot{q}_a - L \quad \Rightarrow \quad L = \sum_a p_a \dot{q}_a - H$$

► Can get for action $S \equiv \int L dt$:

$$\begin{aligned} S &= \int dt \left(\sum_a p_a \dot{q}_a - H(p_a, q_a, t) \right) = \int \left(\sum_a p_a dq_a - H(p_a, q_a, t) dt \right) \\ &= \int \left(- \sum_a q_a dp_a - H(p_a, q_a, t) dt \right) \end{aligned}$$

We dropped surface term $\int d \sum_a (q_a p_a)$ -assume extremes are fixed, so it won't affect equations of motion

Modified Hamilton's principle

$$S = \int \left(\sum_a p_a dq_a - H(p_a, q_a, t) dt \right) = \int \left(- \sum_a q_a dp_a - H(p_a, q_a, t) dt \right)$$

Minimal action principle

On trajectory corresponding to real motion, the action should achieve a minimum

► Now we treat variables p , q as independent and need to repeat what we did when derived Euler-Lagrange equations:

$$\delta \int p_a dq_a = \int \delta p_a dq_a + \int p_a d\delta q_a = \int \delta p_a dq_a - \int \delta q_a dp_a$$

$$\delta H(p_a, q_a, t) = \sum_a \left(\frac{\partial H}{\partial p_a} \delta p_a + \frac{\partial H}{\partial q_a} \delta q_a \right)$$

– Since δp_a and δq_a are independent arbitrary variations, we require that separately coefficients in front of them are zero

$$\Rightarrow dq_a = \frac{\partial H}{\partial p_a} dt, \quad dp_a = - \frac{\partial H}{\partial q_a} dt.$$

which are equivalent to canonical equations.

Control question

In derivation of the canonical equations from the minimal action principle, we always assumed that all variables $\{p_a, q_a\}$ are independent. Analyze how the canonical equations will change in presence of the non-holonomic constraints

$$\psi_a(q_1, \dots, q_n; p_1, \dots, p_k, t) = 0$$

Control question (reply)

As always, introduce constraints into action using the method of Lagrange multipliers (λ_a)

$$\begin{aligned} S &= \int \left(\sum_a p_a dq_a - H(p_a, q_a, t) dt + \sum_n \lambda_n \psi_n dt \right) = \\ &= \int \left(- \sum_a q_a dp_a - H(p_a, q_a, t) dt + \sum_n \lambda_n \psi_n dt \right) \end{aligned}$$

$$\dot{p}_a = - \frac{\partial H(p, q)}{\partial q_a} + \sum_n \lambda_n \frac{\partial \psi_n}{\partial q_a},$$

$$\dot{q}_a = \frac{\partial H(p, q)}{\partial p_a} - \sum_n \lambda_n \frac{\partial \psi_n}{\partial p_a}.$$

- We don't insert "forces" by hand, we insert constraints into minimal action and take derivatives.
- There are additional contribution in equation for $\dot{\mathbf{q}}$ and $\dot{\mathbf{p}}$
- If the constraint is holonomic (does not depend on $\mathbf{p}$), then $\partial \psi / \partial p_a = 0$ and only have contribution in equation for $\dot{p}_a$ ("force")

Restricted forms of Hamilton's principle

Modified Hamilton's principle

- Canonical equations might be obtained from condition that functional

$$\tilde{S} = \int dt \left(\sum_a p_a \dot{q}_a - H(p_a, q_a, t) \right)$$

achieves minimum on the trajectory

- *Restricted forms of the minimal action principle were discovered many years before Hamilton. Now they can be viewed as special cases of Hamilton's principle, but at the time of discovery were important contributions to understanding of theoretical mechanics*

Some historical dates:

- ▶ Fermat's principle: ~ 1662
- ▶ Newtonian formulation: ~ 1687
- ▶ Maupertui's principle: ~ 1744
- ▶ Lagrangian formulation: ~ 1788
- ▶ Hamiltonian formulation: ~ 1833

Maupertui's principle

If we restrict variations in configuration space to **trajectories with constant energy** ($H = E = \text{const}$)

$$\tilde{S} = \int dt \sum_a p_a \dot{q}_a = 2 \int dt T = \min$$

Single particle case:

$$T = \frac{1}{2} \left(\frac{d\mathbf{r}}{dt} \right)^2 \Rightarrow dt = d|r|/\sqrt{2T}$$

$$\tilde{S} = \int d|r| \sqrt{2T} = \int d|r| \sqrt{E - U(\vec{r})}$$

Fermat's principle

For free particles $T = E = \text{const}$, so particle follows the path of the shortest time

$$\tilde{S} = 2E \int_{t_1}^{t_2} dt = 2E(t_2 - t_1) = \min$$

Hamiltonian as generator

Assume that a particle moves in phase space along the trajectory $\{p_a(t), q_a(t)\}$. If we are interested how an arbitrary function $f(p, q, t)$ changes along the trajectory, should take into account both $\partial_t f$ and implicit time-dependence “hidden” in $p(t), q(t)$.

–Using canonical equations, can get

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_a \left(\frac{\partial f}{\partial q_a} \dot{q}_a + \frac{\partial f}{\partial p_a} \dot{p}_a \right) = \frac{\partial f}{\partial t} + \sum_a \left(\frac{\partial H}{\partial p_a} \frac{\partial f}{\partial q_a} - \frac{\partial H}{\partial q_a} \frac{\partial f}{\partial p_a} \right) = \frac{\partial f}{\partial t} + [H, f]$$

$$[H, f] \equiv \sum_a \left(\frac{\partial H}{\partial p_a} \frac{\partial f}{\partial q_a} - \frac{\partial H}{\partial q_a} \frac{\partial f}{\partial p_a} \right) \quad (= \text{Poisson bracket})$$

Lecture 11:

–for any Noether symmetry $(q, p) \rightarrow (q^\varepsilon, p^\varepsilon)$, change of arbitrary function $f(p, q)$ satisfies

$$\frac{df}{d\varepsilon} = \frac{\partial f}{\partial \varepsilon} + [Q, f], \quad (2)$$

where $Q(p, q)$ is a Noether invariant corresponding to the Noether symmetry.

*Hamiltonian (“energy”) is Noether invariant for a system whose action is invariant under $t \rightarrow t + \varepsilon$, so can see that (1) is a particular case of (2) for $\varepsilon \rightarrow t$, $Q \rightarrow H$

Poisson bracket & its properties

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_a \left(\frac{\partial H}{\partial p_a} \frac{\partial f}{\partial q_a} - \frac{\partial H}{\partial q_a} \frac{\partial f}{\partial p_a} \right) = \frac{\partial f}{\partial t} + [H, f], \quad (1)$$

Poisson bracket $[H, f]$

$$[H, f] \equiv \sum_a \left(\frac{\partial H}{\partial p_a} \frac{\partial f}{\partial q_a} - \frac{\partial H}{\partial q_a} \frac{\partial f}{\partial p_a} \right) \quad (2)$$

- From (1,2) conclude that if $f(p, q) = H(p, q)$, then

$$[H, H] = 0, \quad \frac{dH(p, q, t)}{dt} = \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t},$$

so if L does not depend explicitly on time, then $H = \text{const}$ along the trajectory (energy).

- If $f(p, q)$ is some integral of motion ($df/dt = 0$), and $f(p, q)$ does not depend explicitly on time ($\partial f/\partial t = 0$), then

$$[H, f] \equiv \sum_a \left(\frac{\partial H}{\partial p_a} \frac{\partial f}{\partial q_a} - \frac{\partial H}{\partial q_a} \frac{\partial f}{\partial p_a} \right) = 0.$$

Poisson bracket & its properties

Poisson bracket $[H, f]$

$$[H, f] = \sum_a \left(\frac{\partial H}{\partial p_a} \frac{\partial f}{\partial q_a} - \frac{\partial H}{\partial q_a} \frac{\partial f}{\partial p_a} \right) \quad (1)$$

We can always use definition (1) to find Poisson bracket (e.g. just write short script in *Mathematica*). However, this is not very optimal method when we make evaluations manually: frequently many terms in the right-hand side of (1) vanish.

Now we'll try to formulate some simple (yet very powerful) rules which allow to simplify evaluations

Poisson bracket & its properties

$$[H, f] = \sum_a \left(\frac{\partial H}{\partial p_a} \frac{\partial f}{\partial q_a} - \frac{\partial H}{\partial q_a} \frac{\partial f}{\partial p_a} \right) \quad (1)$$

Properties of the Poisson bracket

(below $c_i = \text{const}$, f_i, g are some functions, q_a are coordinates, p_a are momenta)

$$[f, g] = -[g, f], \quad [f, c_i] = 0$$

$$[c_1 f_1 + c_2 f_2, g] = c_1 [f_1, g] + c_2 [f_2, g]$$

$$[f_1 f_2, g] = [f_1, g] f_2 + f_1 [f_2, g]$$

$$[f, q_a] = \frac{\partial f}{\partial p_a}, \quad [f, p_a] = -\frac{\partial f}{\partial q_a},$$

$$[q_a, q_b] = 0, \quad [p_a, p_b] = 0, \quad [p_a, q_b] = \delta_{ab},$$

$$[[f, g], h] + [[h, f], g] + [[g, h], f] = 0$$

These properties allow to simplify evaluation of Poisson brackets of complex functions. It is strongly recommended to use them instead of “direct” evaluation from definition (saves a lot of time)

Poisson bracket & its properties

- As we discussed, there is no need to start from definition

$$[f, g] = \sum_a \left(\frac{\partial f}{\partial p_a} \frac{\partial g}{\partial q_a} - \frac{\partial f}{\partial q_a} \frac{\partial g}{\partial p_a} \right)$$

which includes explicit summation over all derivatives (frequently many terms eventually vanish)

- The relations

$$[f, q_a] = \frac{\partial f}{\partial p_a}, \quad [f, p_a] = -\frac{\partial f}{\partial q_a},$$

suggest that eventually Poisson brackets with coordinates and momenta behave like derivative operators

- ▷ Assume $f(p)$ and $g(q)$ are two functions of momenta p and coordinates q respectively. Then

$$[f, g] = \frac{\partial f(p)}{\partial p} \frac{\partial g(q)}{\partial q}$$

Basic examples

$$[\sin p, q] = \frac{\partial \sin p}{\partial p} = \cos p,$$

$$[\sin p, p^2] = 2p [\sin p, p] = -2p \frac{\partial \sin p}{\partial q} = 0,$$

$$[\sin p, q^2] = 2q [\sin p, q] = 2q \frac{\partial \sin p}{\partial p} = 2q \cos p,$$

$$\begin{aligned} [\sin p, p^2 + 2q^2] &= [\sin p, p^2] + 2 [\sin p, q^2] \\ &= 4q \cos p \end{aligned}$$

–2D example: assume q_1, q_2 are coordinates and p_1, p_2 are momenta:

$$[\sin p_1, q_1] = \frac{\partial \sin p_1}{\partial p_1} = \cos p_1,$$

$$[\sin p_1, q_2] = \frac{\partial \sin p_1}{\partial p_2} = 0,$$

More sophisticated examples

Assume that q_a are generalized coordinates, p_a are corresponding momenta, $M_a = \epsilon_{abc} q_b p_c$ are components of the angular momentum, and $\mathbf{M}^2 = \sum_a M_a M_a$

$$[M_a, p_b] = [\epsilon_{aij} q_i p_j, p_b] = \epsilon_{aij} \left(\underbrace{[q_i, p_b]}_{-\delta_{ib}} p_j + q_i \underbrace{[p_j, p_b]}_{=0} \right) = -\epsilon_{abj} p_j$$

$$[M_a, q_b] = [\epsilon_{aij} q_i p_j, q_b] = \epsilon_{aij} \left(\underbrace{[q_i, q_b]}_{=0} p_j + q_i \underbrace{[p_j, q_b]}_{=\delta_{jb}} \right) \epsilon_{ajb} q_j = -\epsilon_{abj} q_j$$

$$\begin{aligned} [M_a, M_b] &= [\epsilon_{aij} q_i p_j, M_b] = \epsilon_{aij} \left(\underbrace{[q_i, M_b]}_{-\epsilon_{ibk} q_k} p_j + q_i \underbrace{[p_j, M_b]}_{-\epsilon_{jbk} p_k} \right) \\ &= -\epsilon_{aij} \epsilon_{ibk} q_k p_j - \underbrace{\epsilon_{aij} \epsilon_{jbk} q_i p_k}_{(ijk) \rightarrow (kij)} = \underbrace{(\epsilon_{aij} \epsilon_{ibk} + \epsilon_{aki} \epsilon_{ibj})}_{\epsilon_{abs} \epsilon_{skj}} q_k p_j = -\epsilon_{abs} M_s \equiv -\epsilon_{abc} M_c \end{aligned}$$

$$[M^2, M_z] = 2 \sum_a M_a [M_a, M_z] = -2 \sum_{aj} \epsilon_{azj} M_a M_j = 0 \quad \text{due to } \epsilon_{azj} = -\epsilon_{jza}$$

Example 3

The particle moves in the central field $U(r)$. Demonstrate that all components M_a of angular momentum are integrals of motions

$$\frac{\partial M_a}{\partial t} = 0 \Rightarrow \text{have to show that } [H, M_a] = 0$$

$$H = \frac{p^2}{2m} + U(r), \quad r = \sqrt{x^2 + y^2 + z^2}$$

$$[H, M_z] = \frac{1}{2m} [p^2, M_z] + [U(r), M_z],$$

$$[p^2, M_z] = 2 \sum_a p_a [p_a, M_z] = -2 \sum_{aj} \epsilon_{azj} p_a p_j = 0$$

due to antisymmetry of $\epsilon_{azj} = -\epsilon_{jza}$

$$\begin{aligned} [U(r), M_z] &= [U(r), q_x p_y - q_y p_x] = \\ &= \underbrace{[U(r), q_x]}_{=0} p_y + \underbrace{q_x}_{=x} \underbrace{[U(r), p_y]}_{-\partial U / \partial y} - \underbrace{q_y}_y \underbrace{[U(r), p_x]}_{-\partial U / \partial x} - p_x \underbrace{[q_y, U(r)]}_{=0} = \\ &= -U'(r) \frac{x y}{r} + U'(r) \frac{x y}{r} = 0 \end{aligned} \quad \Rightarrow [H, M_z] = 0.$$

Poisson's theorem

Poisson's theorem

If f, g are integrals of motion, then $h = [f, g]$ is also an integral of motion

- Obviously, for any system number of integrals of motion is finite. After several iterative application of procedure, the “new” integrals of motion h_i either will start vanishing, or reduce to linear combination of known integrals

Lecture 13:

–For any continuous symmetry of action, there is its own integral of motion $Q_a \in \{f, g, \dots\}$.

*These Noether invariants (integrals of motion) play the role of generators of corresponding transformation

$\Rightarrow$ Poisson's theorem = claim that generators form Lie algebra, i.e. the set of transformations which don't change action should form a group

Poisson's theorem (proof)

First we'll assume that f, g do not depend on time

– They are integrals of motion, so $[H, f] = [H, g] = 0$

– Need to show that $[H, h] = 0$ for $h = [f, g]$

Jacobi's identity:

$$[[f, g], h] + [[h, f], g] + [[g, h], f] = 0$$

Replace $h \rightarrow H$ (hamiltonian)

$$\left[\underbrace{[f, g]}_h, H \right] + \left[\underbrace{[H, f]}_{=0}, g \right] + \left[\underbrace{[g, H]}_{=0}, f \right] = 0$$

$\Rightarrow [H, h] = 0$, which finishes the proof for this special case

Poisson's theorem (proof)

- Generalize for the case when f, g depend on time
 - They are integrals of motion, so

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + [H, f] = 0, \quad \frac{dg}{dt} = \frac{\partial g}{\partial t} + [H, g] = 0$$

- Need to show that $dh/dt = \partial_t h + [H, h] = 0$

$$\begin{aligned} \frac{dh}{dt} &= \frac{d}{dt} [f, g] = \left[\frac{\partial f}{\partial t}, g \right] + \left[f, \frac{\partial g}{\partial t} \right] + [H, [f, g]] = \\ &= \left[\frac{\partial f}{\partial t}, g \right] + \left[f, \frac{\partial g}{\partial t} \right] + [[H, f], g] + [f, [H, g]] = \\ &= \left[\frac{df}{dt}, g \right] + \left[f, \frac{dg}{dt} \right] = 0 \end{aligned}$$

Q.E.D.

Examples

Assume that a particle moves in some potential field. Demonstrate that:

- 1) If M_x and M_y are integrals of motion, then M_z also must be integral of motion
- 2) If M_x and p_y are integrals of motion, then p_z also must be integral of motion
- 3) If M_x , M_y and p_z are integrals of motion, then all the remaining components M_z , p_x , p_y also must be integrals of motion
- 4) As You remember, Noether theorem relates continuous symmetries of the system (its action) with some integrals of motion. Recall to what symmetries correspond conservation of momentum $\mathbf{p}$ and angular momentum $\mathbf{M}$. Reformulate the results of items (1-3) in terms of the symmetries of the system.